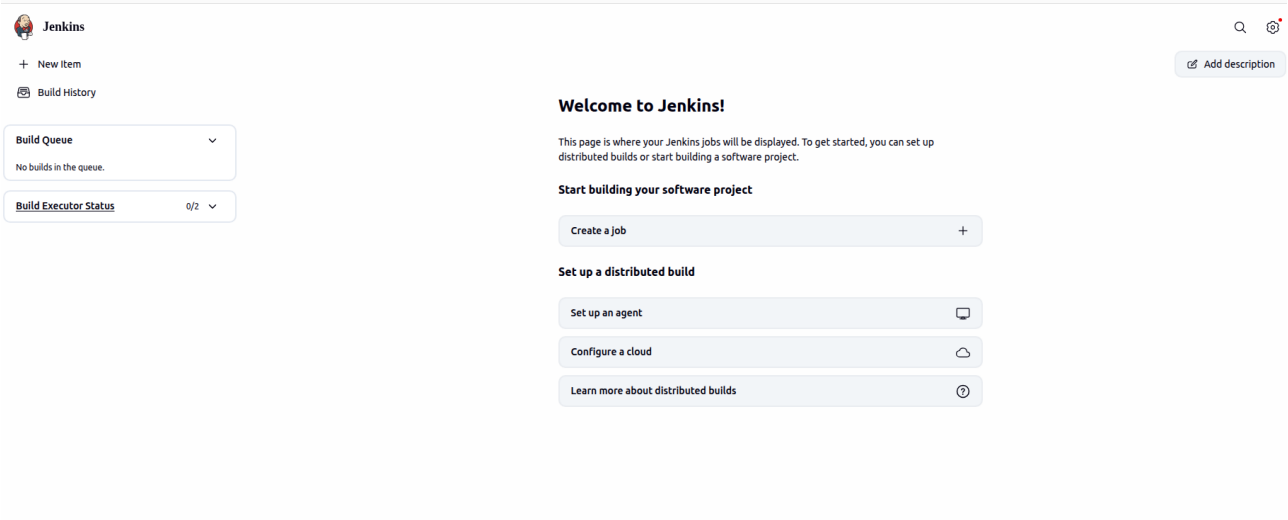
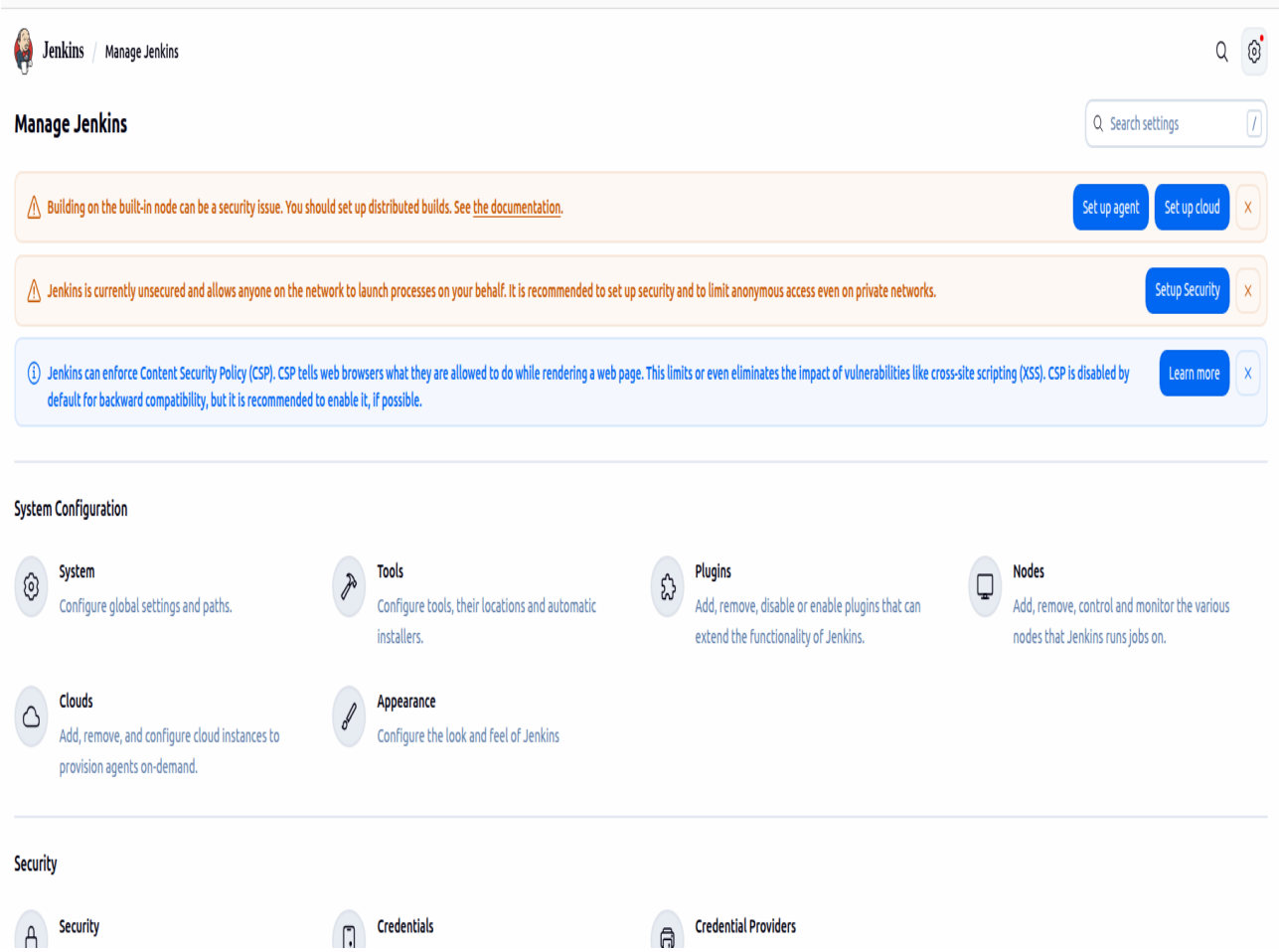
Program 8: Jenkins Plugins Installation,Tolls Configuration & Pipleline setup for simple maven

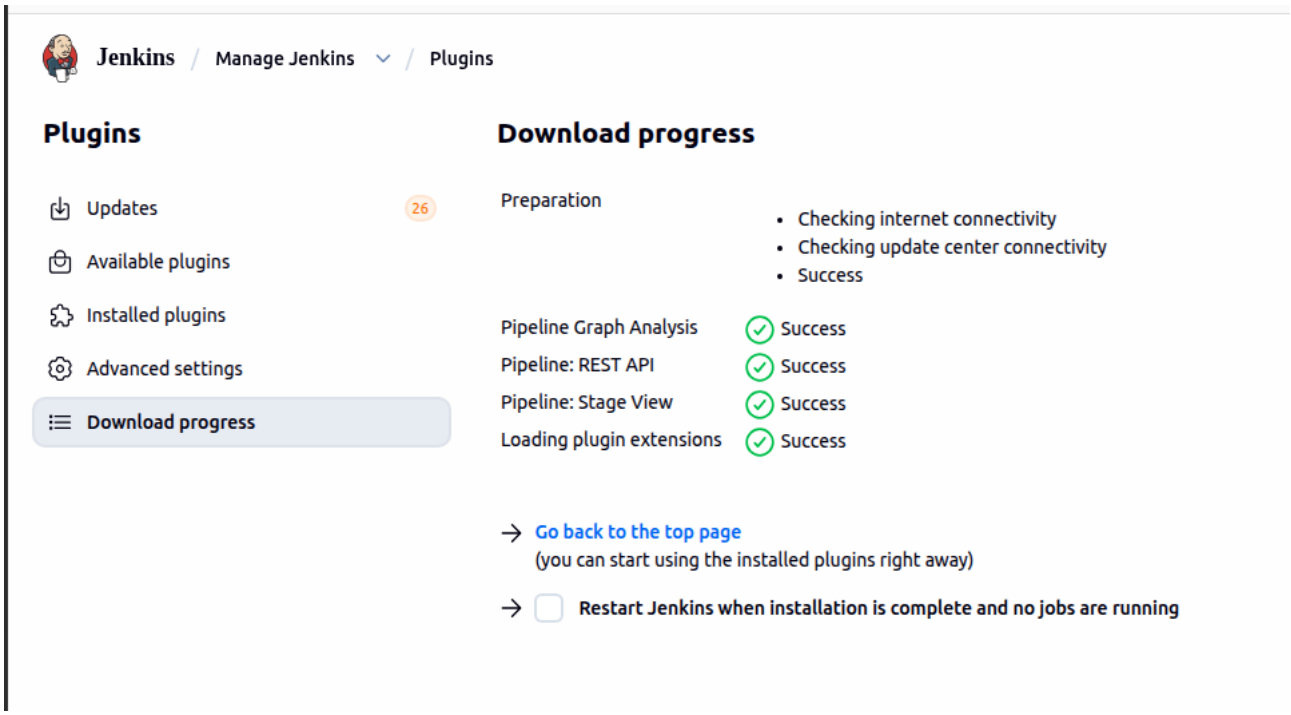
Step 1: Login to Jenkins

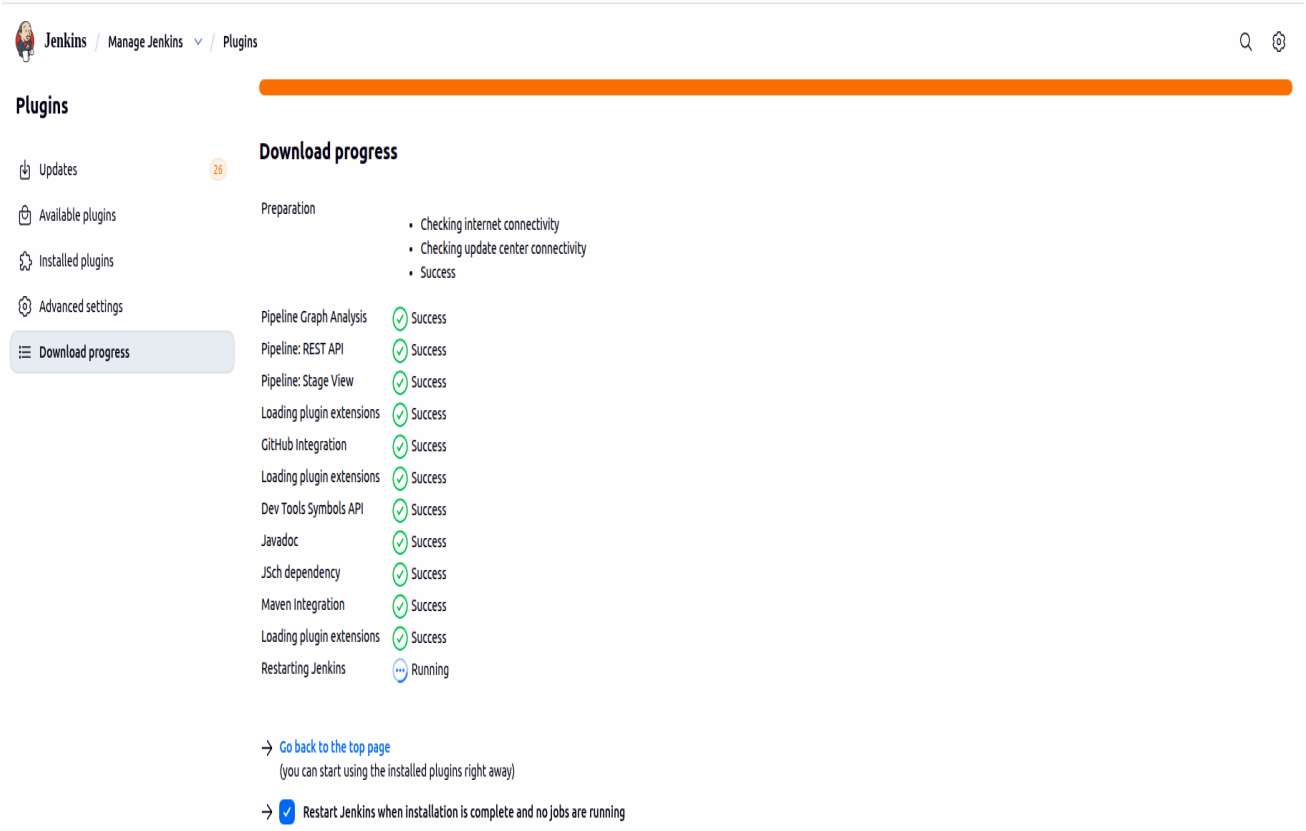


Step 2 : Go to Manage Jenkins

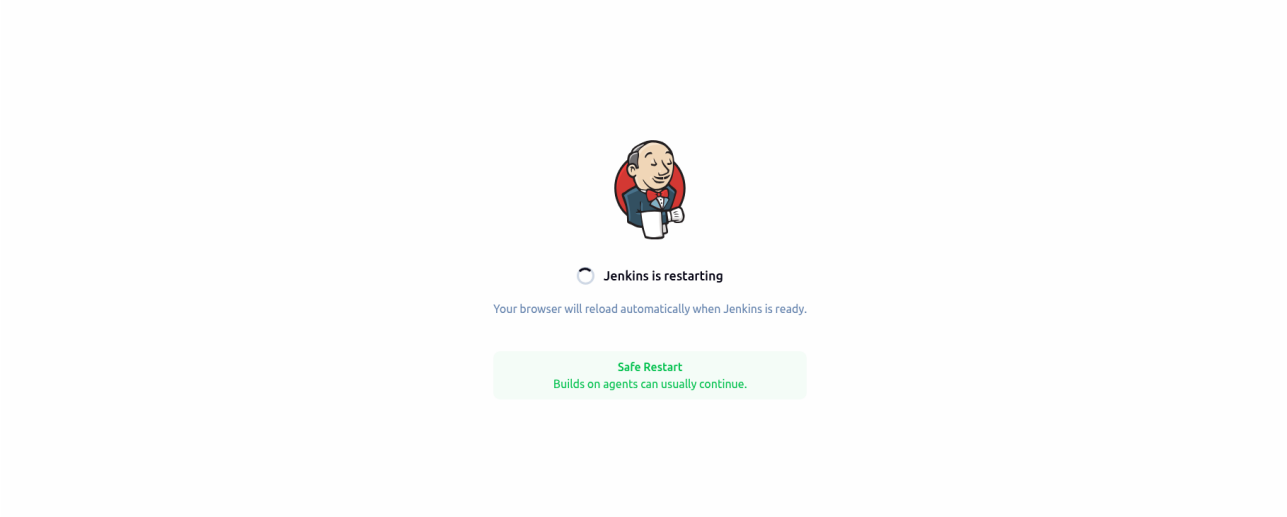


Step 3: Click on plugins and install the required plugins
1.Pipleline Stage view
2.Github Integration
3.Maven
4.Gradle

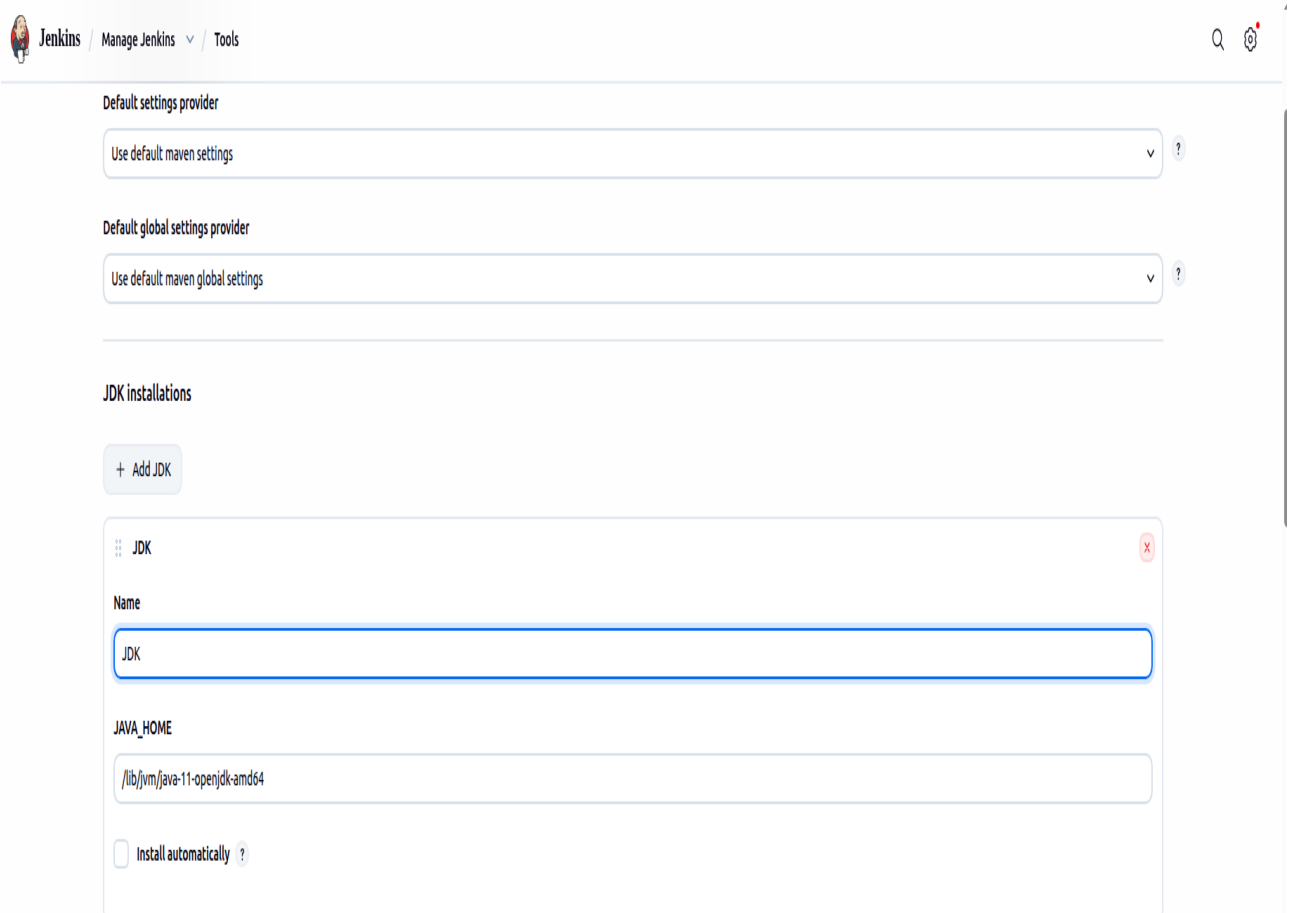




Step 4: After verifying installation restart jenkins

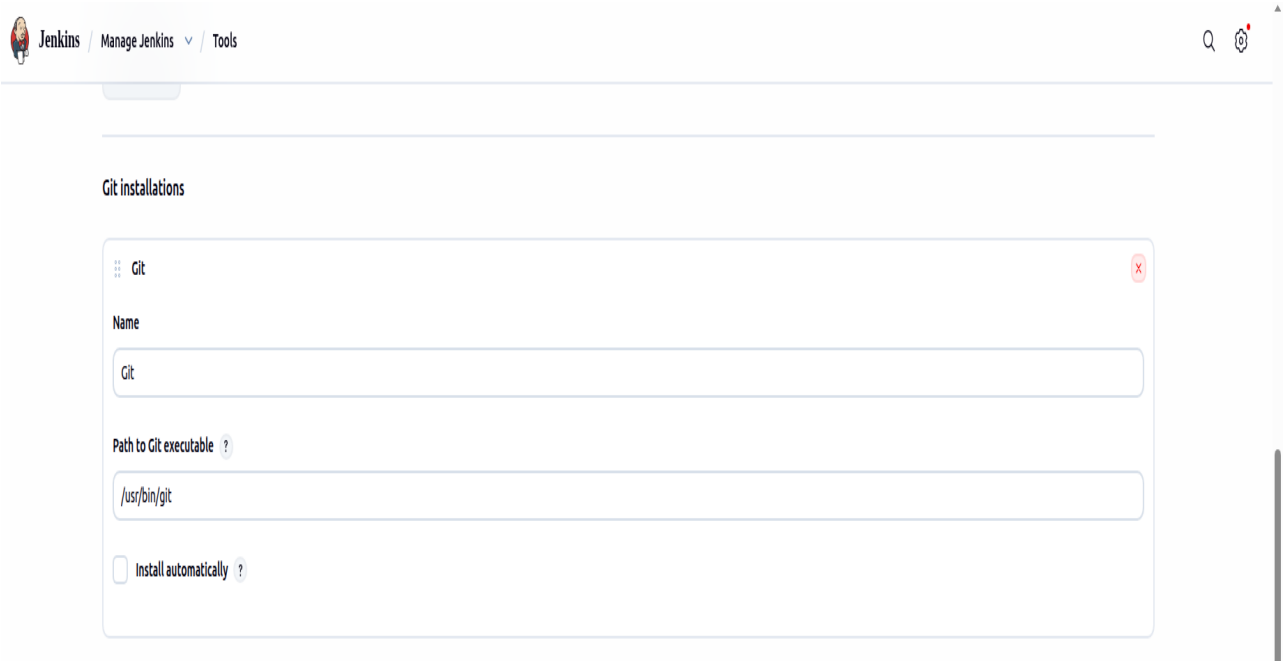


Step 5: Configure Tools
Go to Manage Jenkins → Tools.
Scroll down and click Add JDK.
Enter JDK name and path (Computer → lib → jvm → jdk).

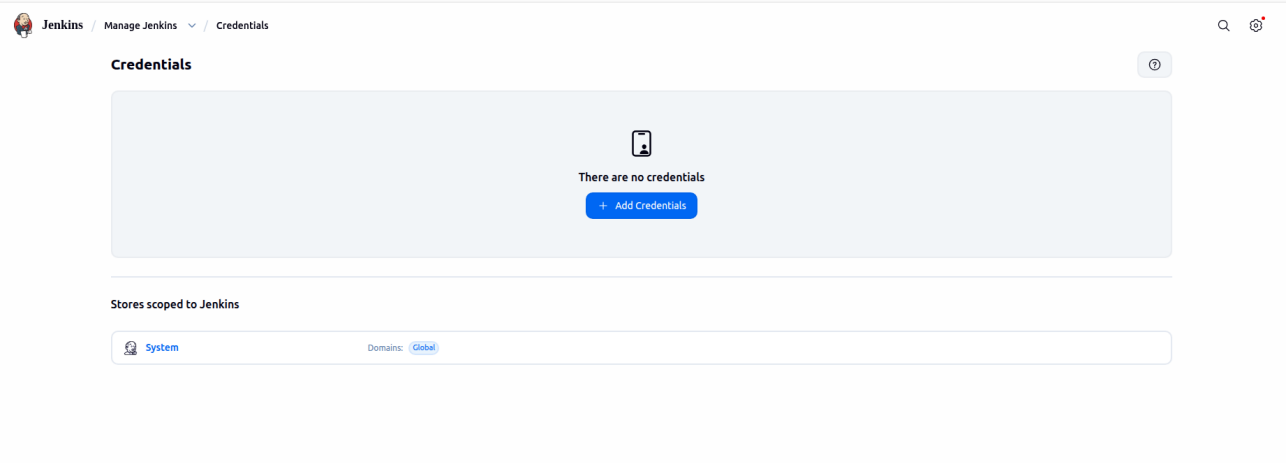


Enter the name and address The address of the JDK can be found in your files -> other location -> computer -> lib -> jvm -> jdk

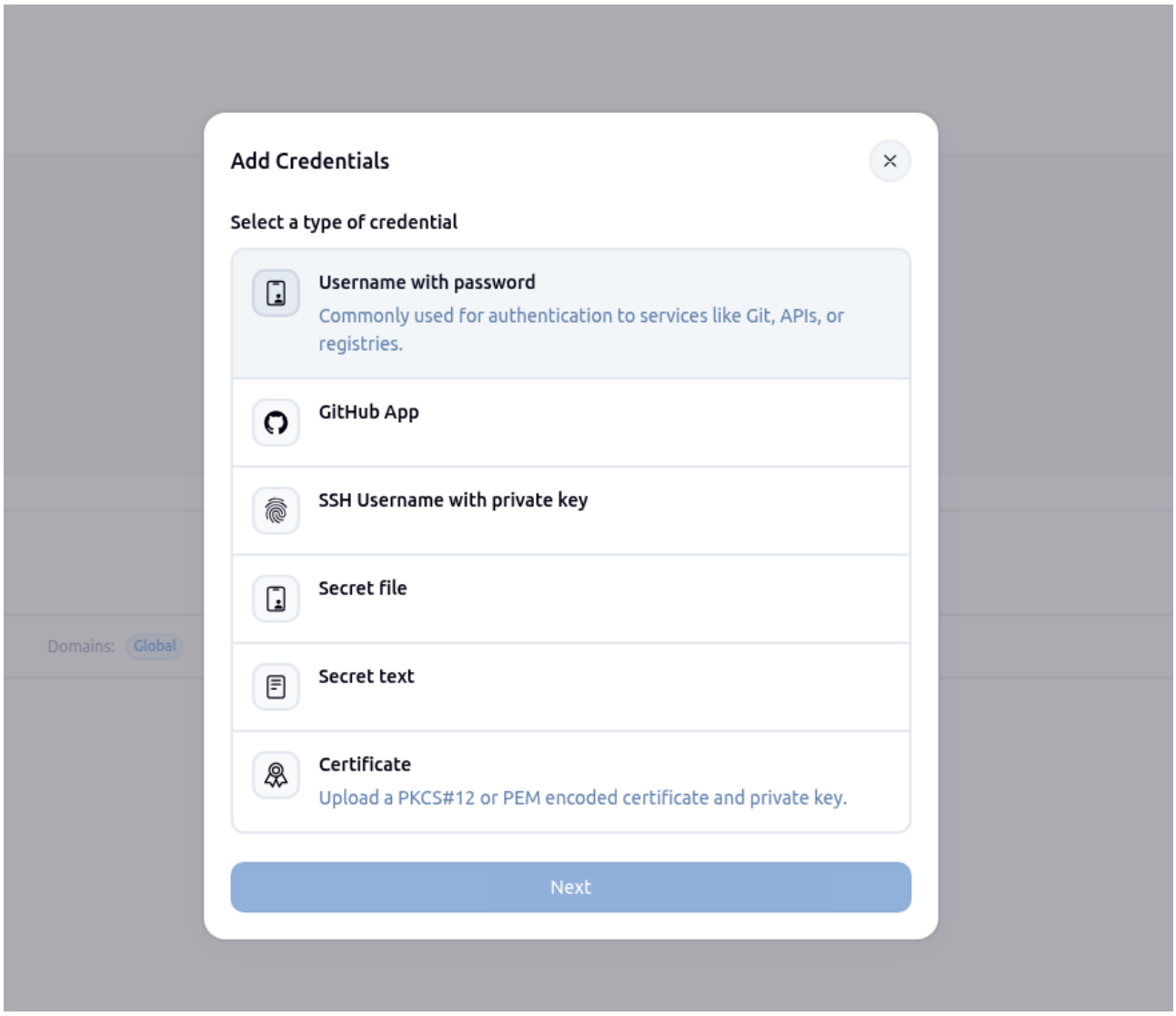
Step 7: Configure Build Tools Find installation paths for Git, Maven, and Gradle using terminal.
Enter these paths in Jenkins and click Apply and Save.



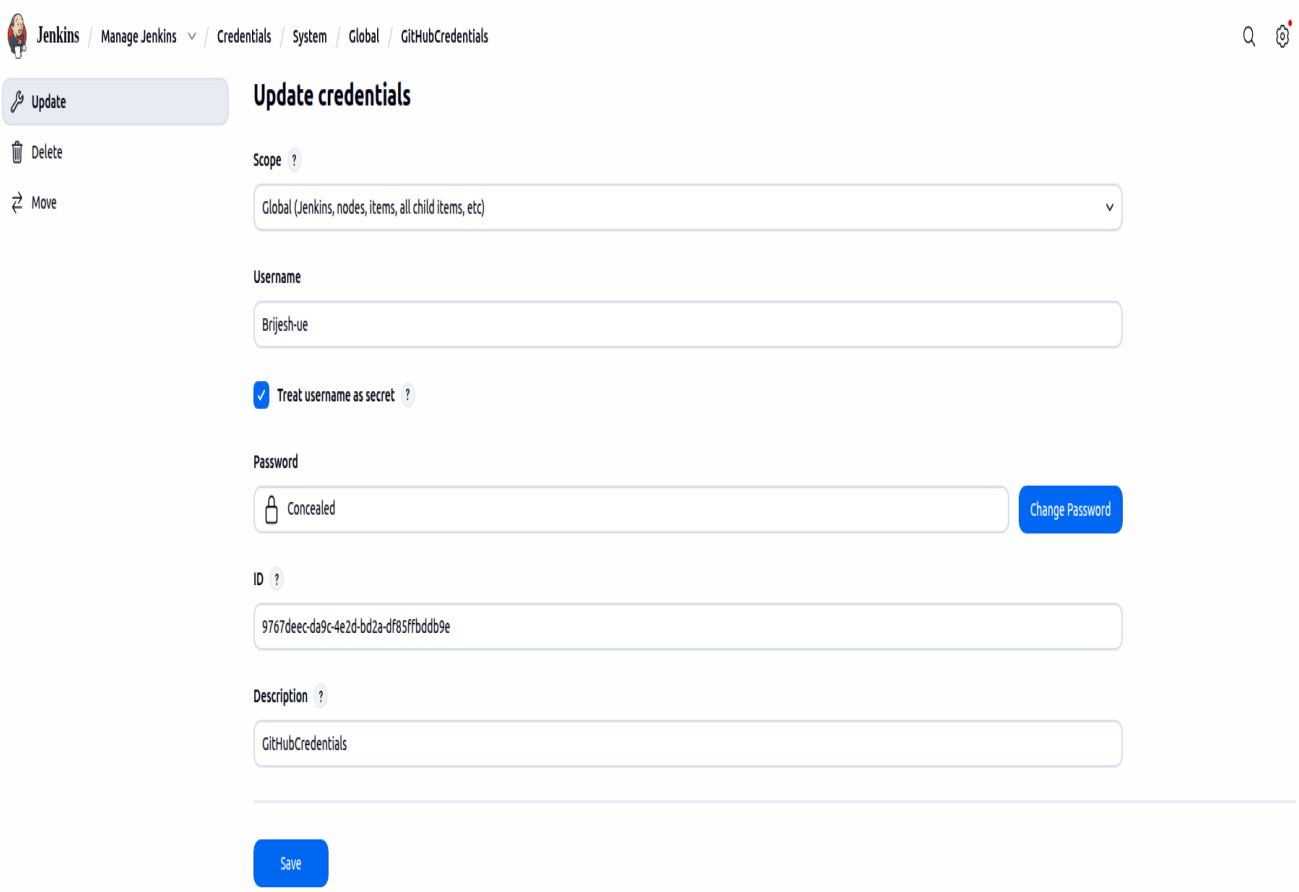
Step 8: Add GitHub Credentials
Go to Manage Jenkins → Credentials



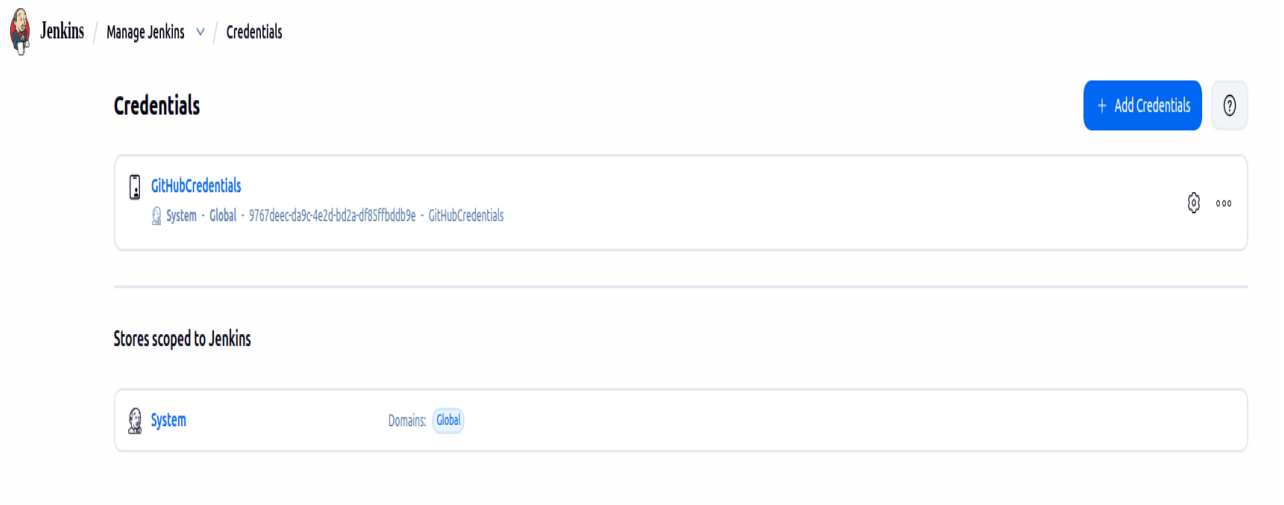
Add Username with Password and click on next



Enter GitHub credentials and description as 'GitHub Credentials'.

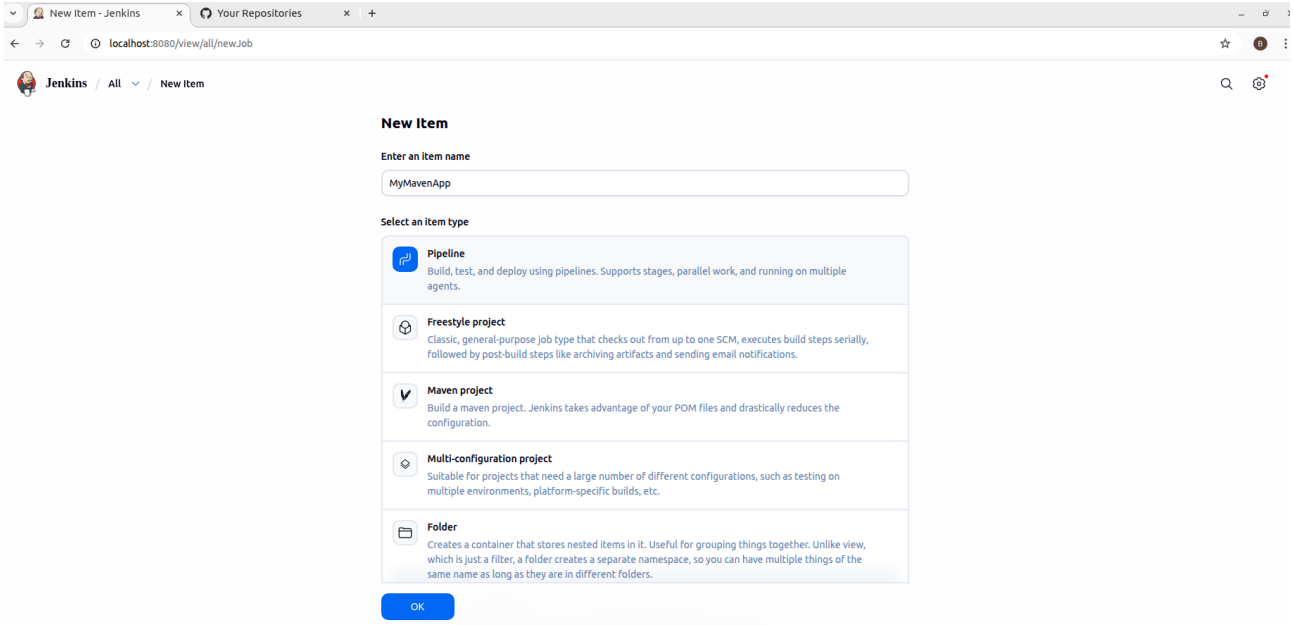


Verify if we have successfully created the Credential



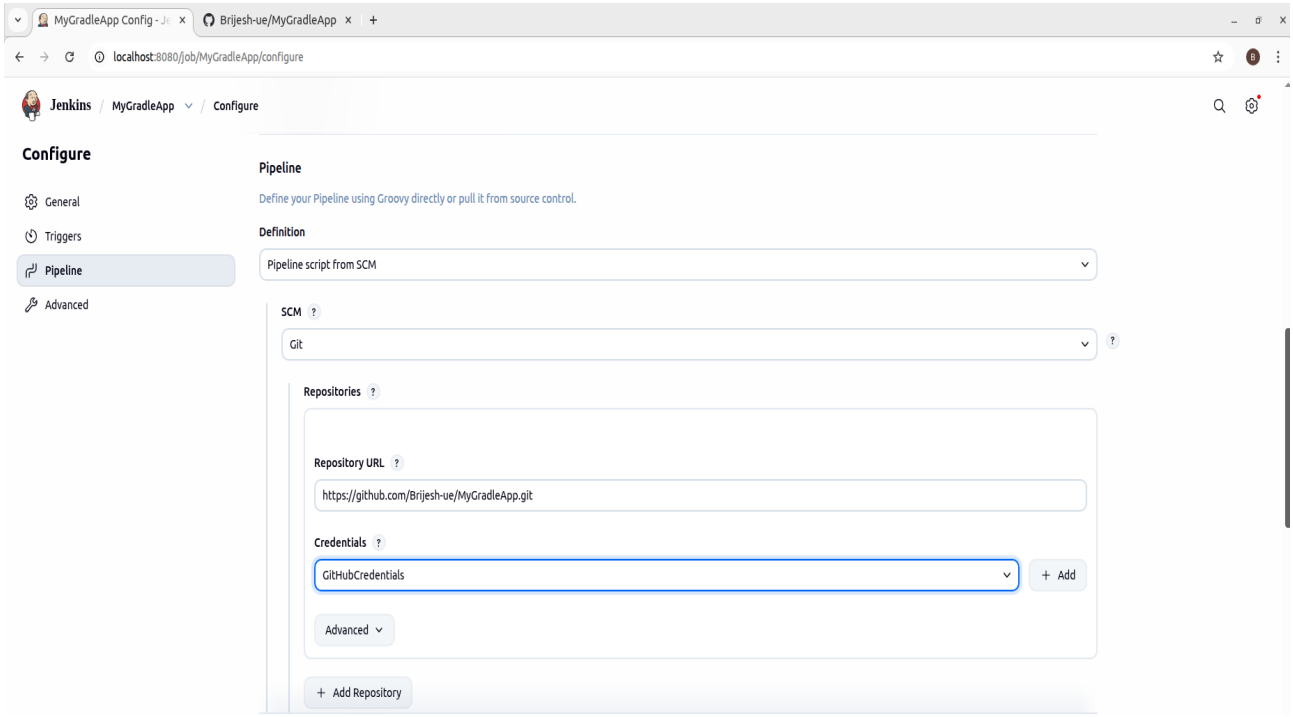
Step 9: Create a New Pipeline Job

Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.



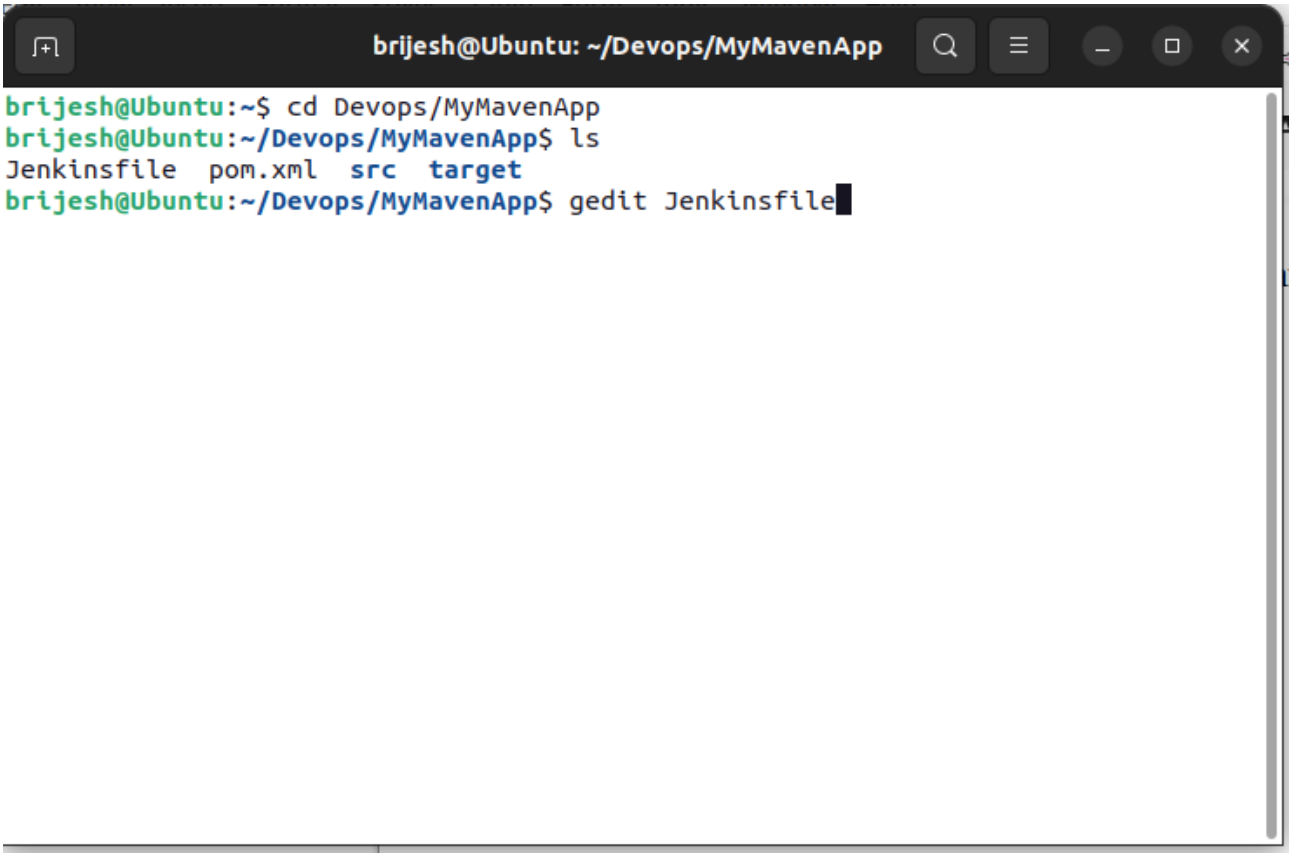
Step 10: Configure Pipeline from SCM

Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.
Click Apply and Save.



Step 11: Create Jenkinsfile

Create a Jenkinsfile in your MyMavenApp project and add the pipeline script.



Enter following code into jenkins file

```
pipeline {
    agent any  // Use any available agent

    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url:
'https://github.com/Brijesh-ue/Mymavenapp.git'
            }
        }
```

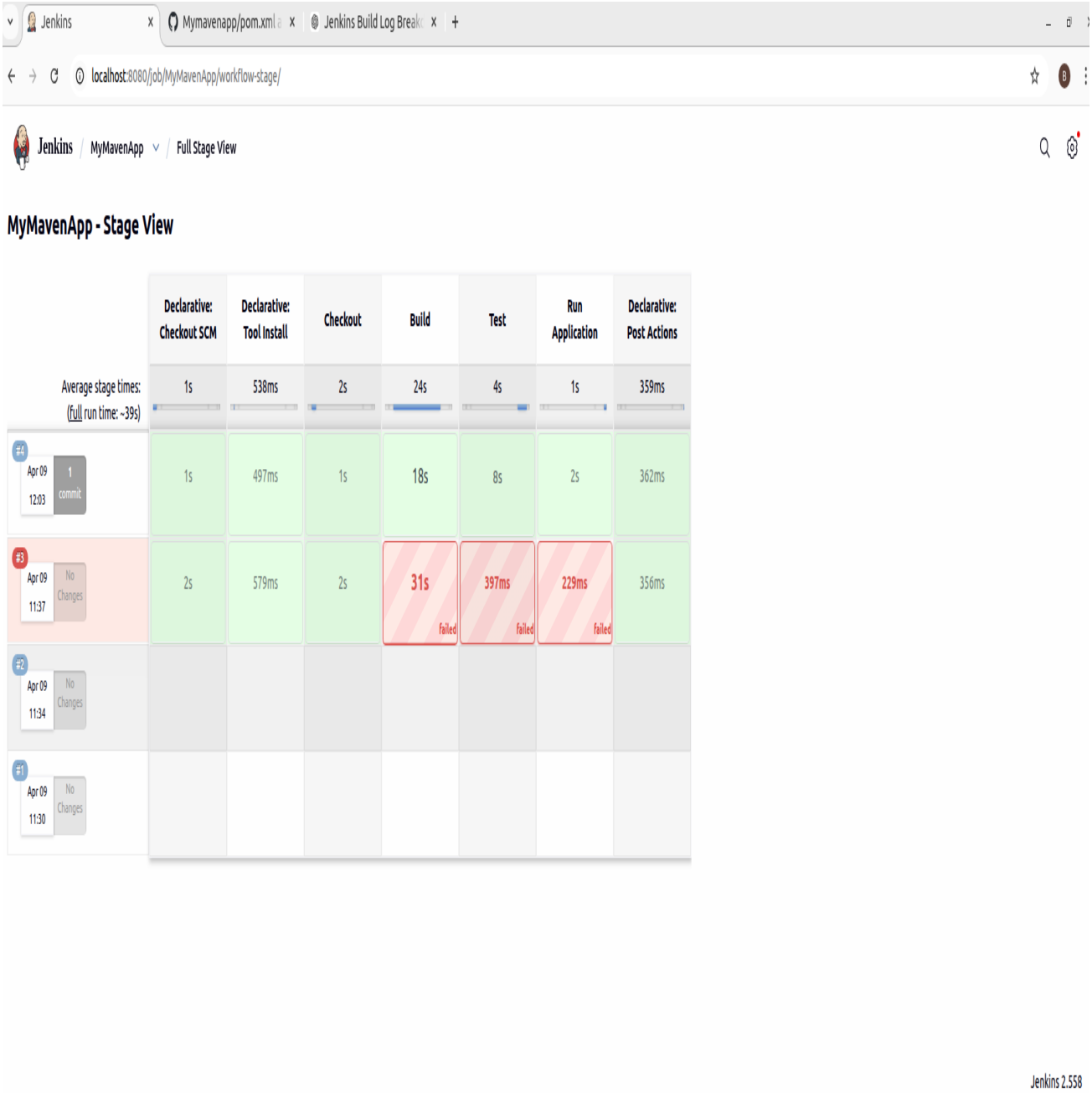
```
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }




        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
            }
        }


    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 12: Update pom.xml

Change source and target versions from 1.7 to 1.8.
Commit changes and push to your github repository.

Step 13: Build the Pipeline

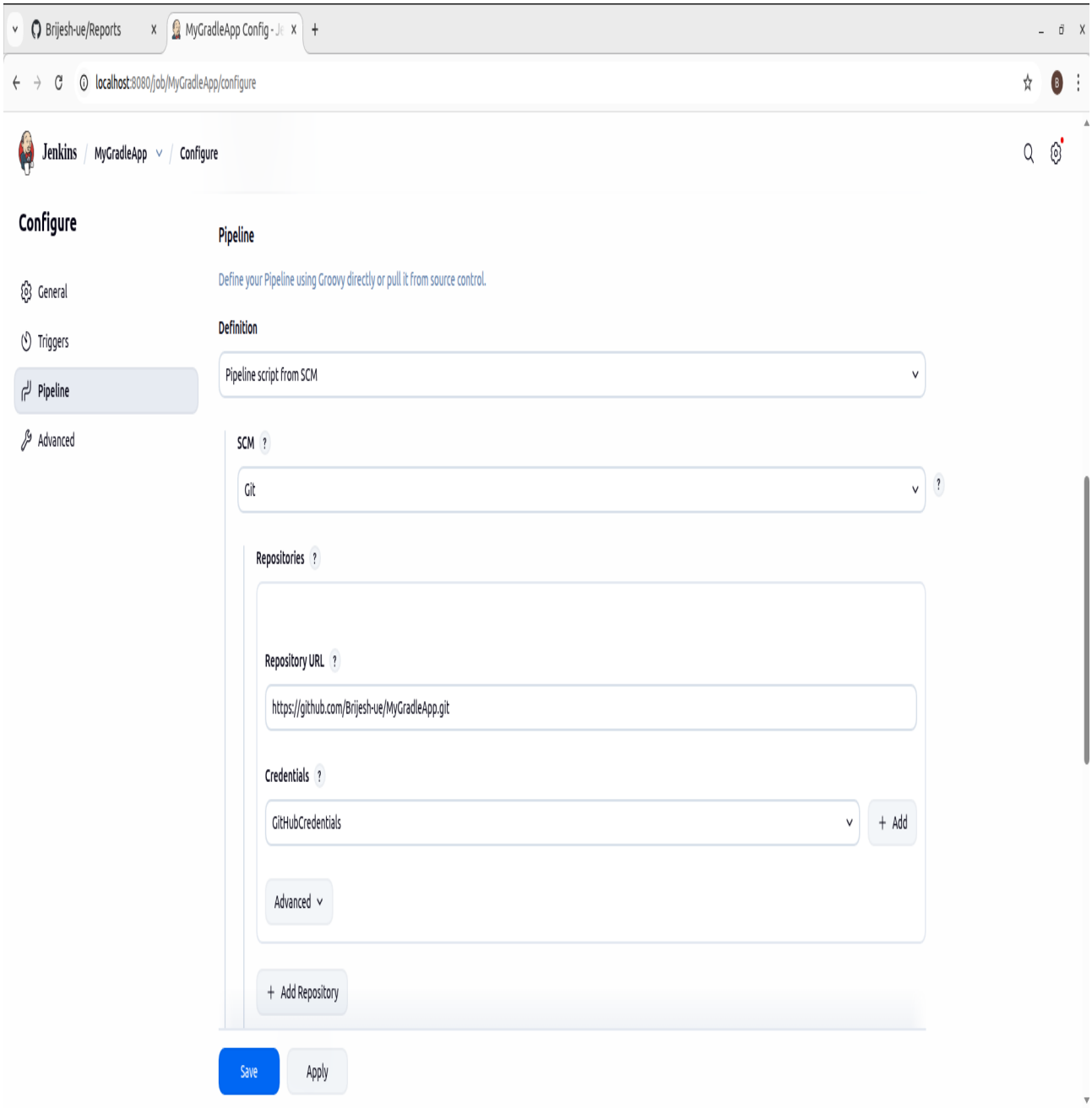
Go to Jenkins and click Build Now.
Monitor Build Stages Use Stage View and Full Stage View to monitor execution.



Gradle pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.

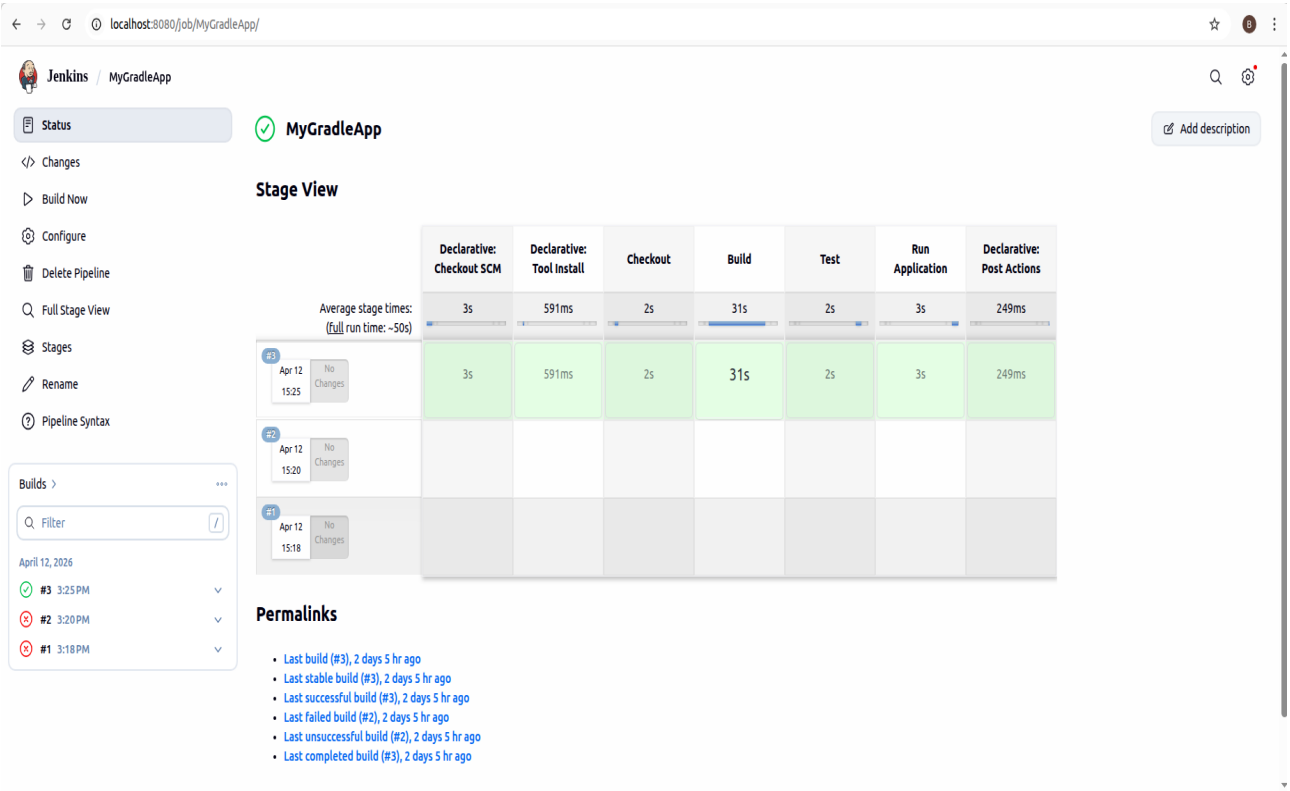


Step 3: Create Jenkinsfile
Create a Jenkinsfile in your MyGradleApp project and add the pipeline script.

```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyGradleApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }




        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
```

```
        }
    }


}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

Monitor Build Stages Use Stage View and Full Stage View to monitor execution.